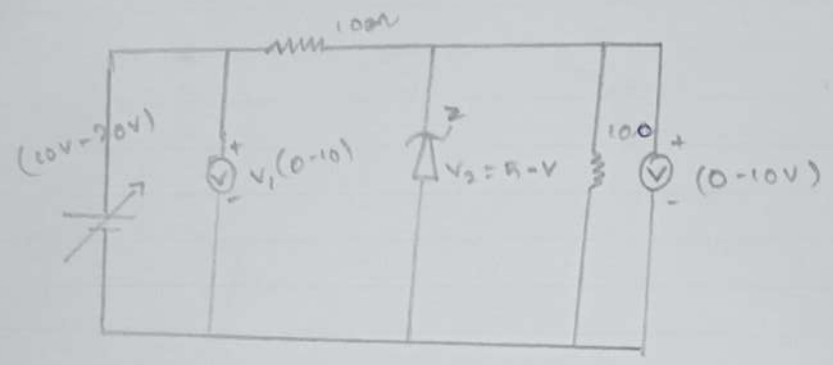
Circuit Diagram:-

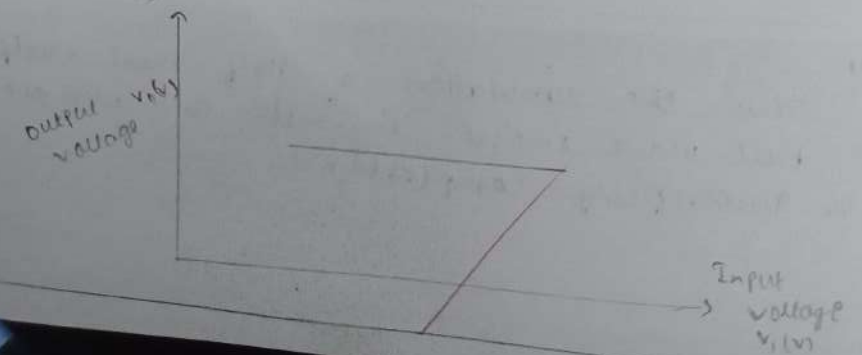
Line regulation.



Tabulation

S.NO.	Input voltage V_i(V)	Output voltage V_o(V)
1	+12.0	+5.14
2	+13.0	+5.16
3	+14.0	+5.17
4	+15.0	+5.17
5	+16.0	+5.20
6	+17.0	+5.21

Model Graph:



zener diode is as voltage Regulation.

Aim:

To simulate the line and load regulation operation zener diode.

Apparatus Required:

Laptops with proteus software.

Theory:

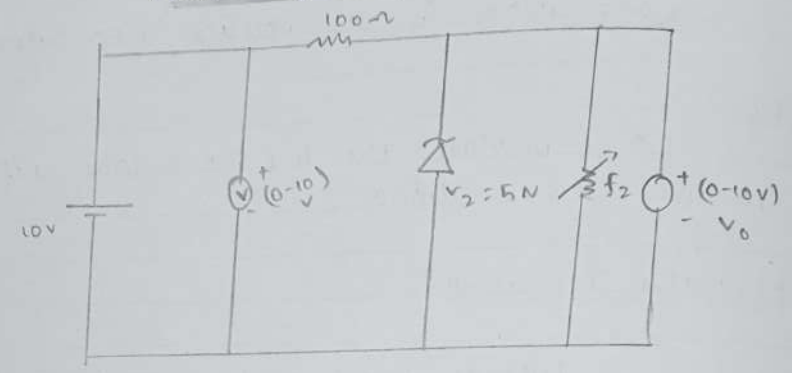
A heavily dopped PN junction diode is called zener diode. Due to heavily dopped nature zener diode works under forward bias and reverse bias conditions.

zener diode under reverse bias condition is used as a voltage regulation. Even if the input voltage changes the output is constant. This is called line regulation.

If the load resistance changes the output voltage is constant this is called load regulation.

Circuit Diagram:

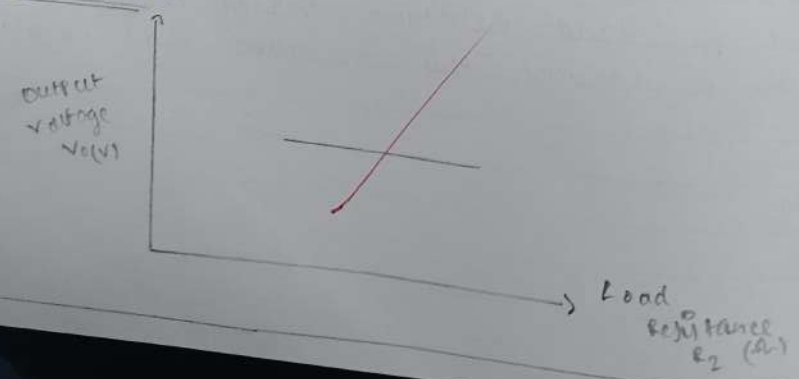
Load Regulation:



Tabulation:

S.NO	Load Resistance $R_2(\Omega)$	Output voltage V_0 (V)
1	100 Ω	+5.19
2	90 Ω	+5.19
3	80 Ω	+5.18
4	70 Ω	+5.17
5	60 Ω	+5.16
6	50 Ω	+5.15

Model Graph:

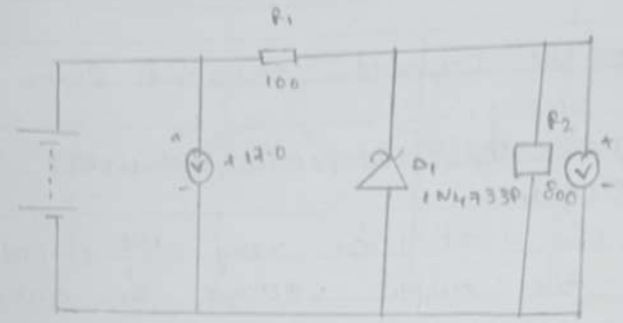


Procedure:

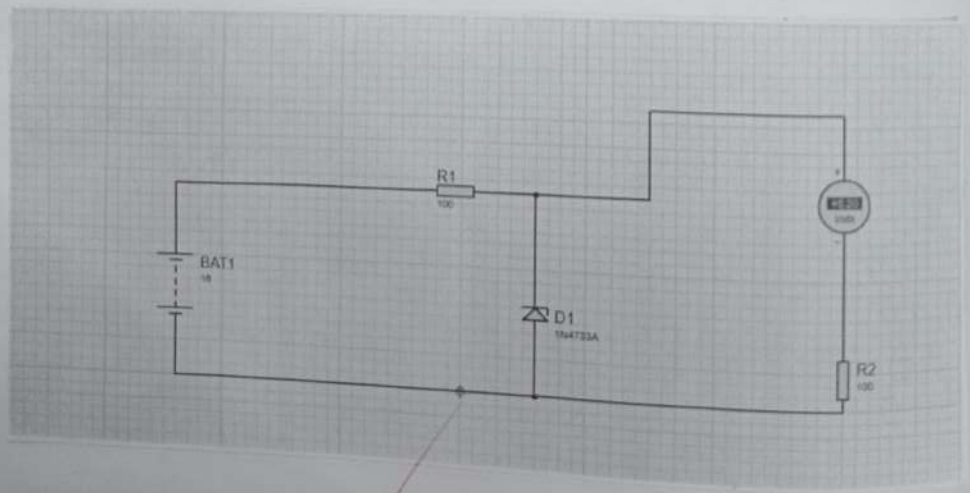
1. Drag the required components from the Protem library.
2. Connect the components (required) as per circuit diagram.
3. For line regulation vary the input voltage and note the output voltages in tabulation. Draw the graph between input voltage and output voltage.
4. For Load regulation vary the load resistance and note the output voltage in tabulated. Draw the Graph between load resistance and output voltage.

Line Regulation :

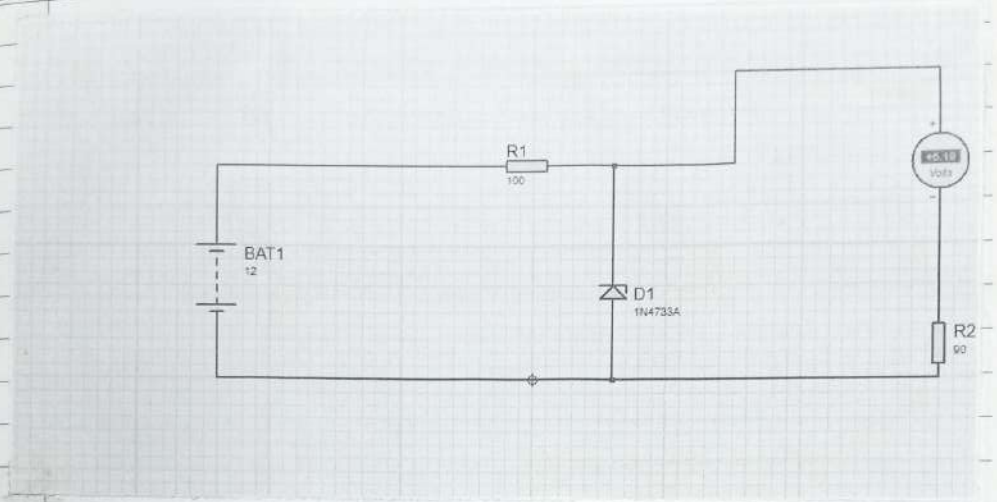
output :



Line :



Load:



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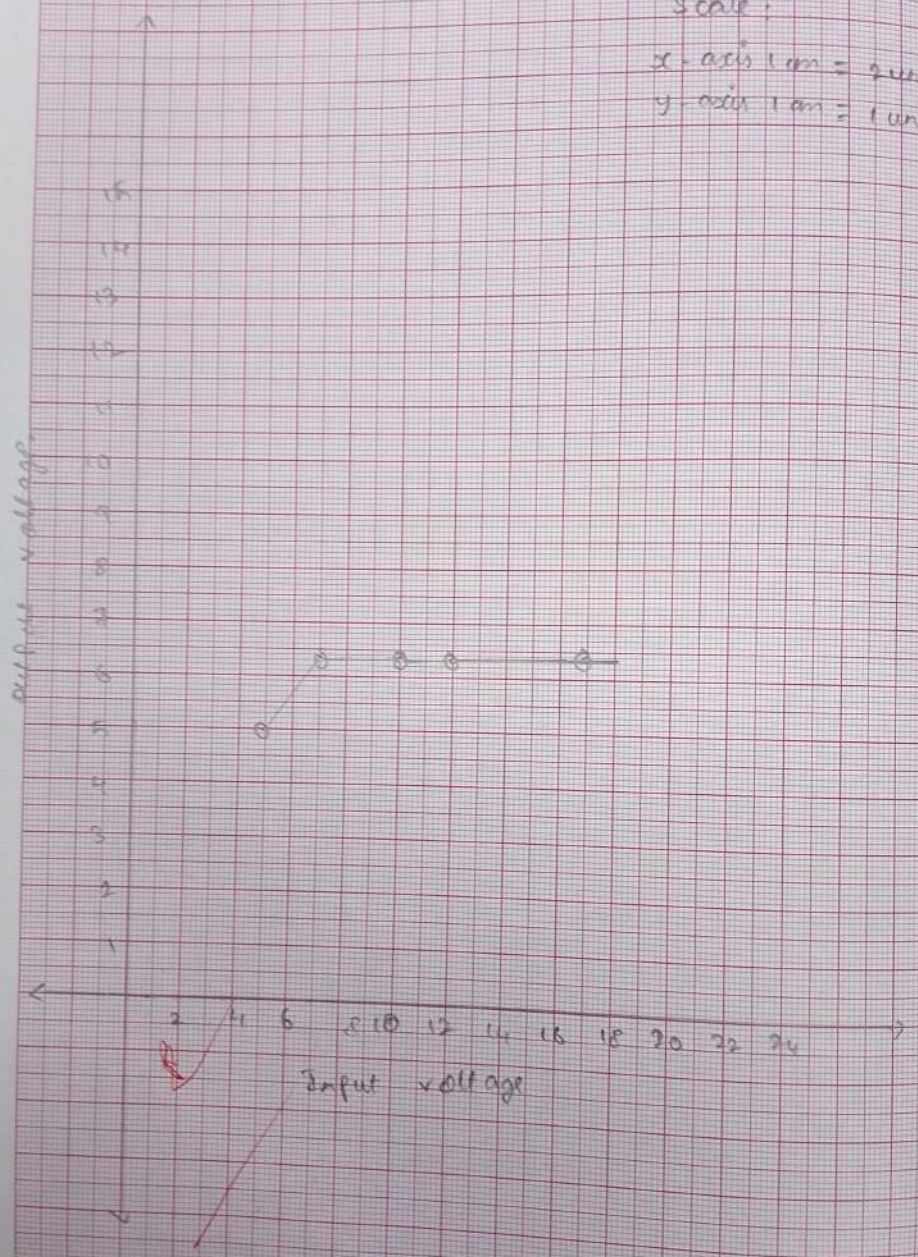
Result:

Thus simulation of line load regulation operation of zener diode successfully completed.

Scale :

x - axis 1 cm = 2 unit

y - axis 1 cm = 1 unit



Expt. No.

Page No.

zener diode as voltage Regulation

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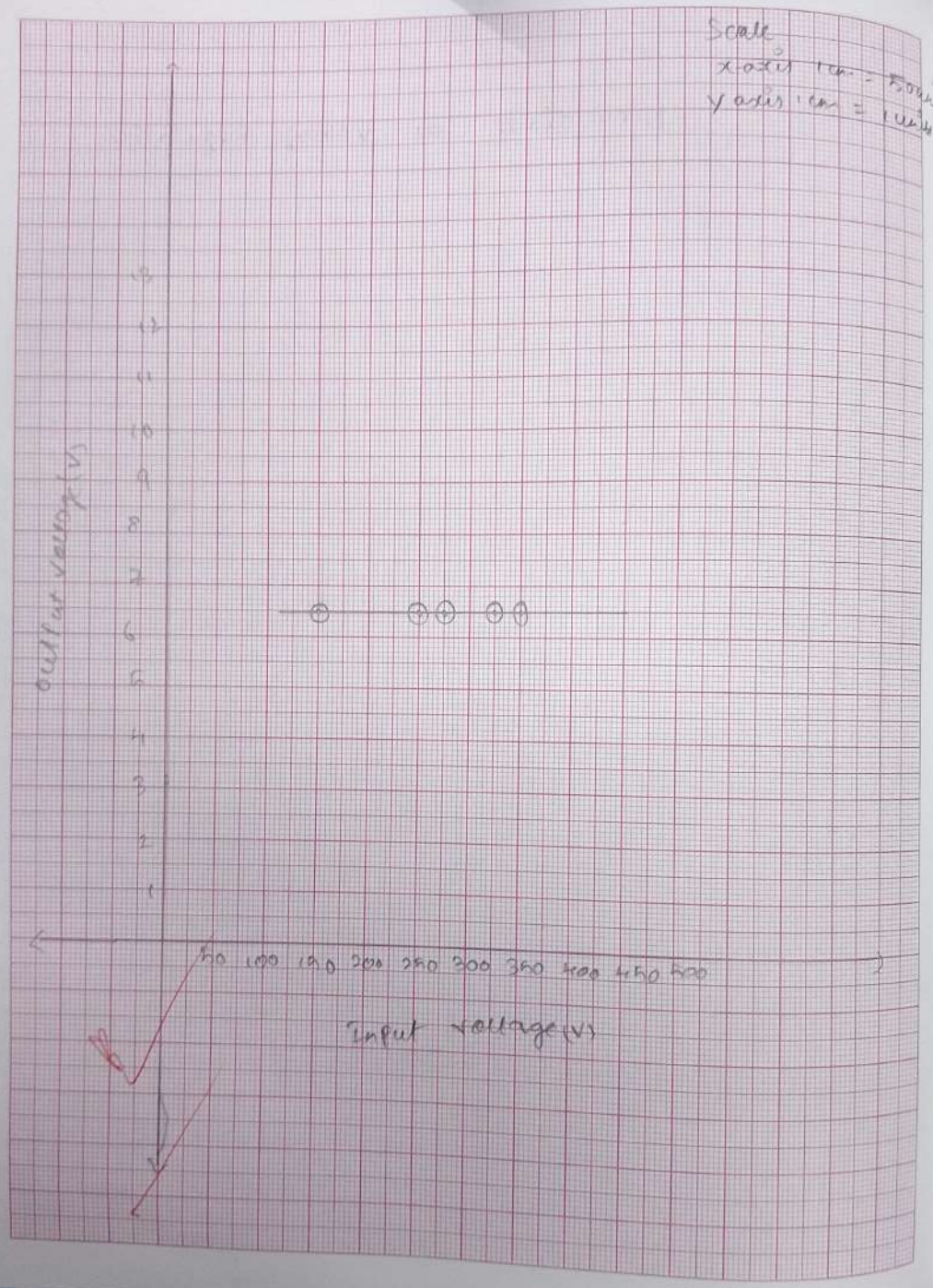
Scale
x axis 1cm = 50 units
y axis 1cm = 1 unit
output voltage (V)
1
2
3
4
5
6
7
8
9
10
11
12
13
50 100 150 200 250 300 350 400 450 500
Input voltage (V)